10/2

Typing this first entry for speed. The results of Assignment 2 helped to make design decisions pertaining to the Wheatstone bridge, amplifier selection and gain resistors, and it is probable that a couple things will need to change in order to acheive our end goal for the purpose of measuring the temperature of our already brewed, hot cup of coffee. Since it was found that the average comfortable drinking range (which is different than the brewing range of 90C to 96C) is from 54C-70C. Since particular people I know actually prefer scalding coffee since their taste buds simply are burnt to the point of no return, a common 25 - 85 degree band seems like it would be fine to work with. Once you hit 70 C, anything above that is most likely undrinkable. So the 15 degree margin seems adequate.

One thing I ran into in the previous design is that my given manufacturing table for the Semitec 103AP-2-A only gives 13 values. This caused some strange results in the Steinhart Hart model I created with it because LTSpice was interpolating between those temperatures. The Steinhart-Hart model is supposed to be more accurate because it uses three calibration points instead of 2. Additionally, trying to plot both Beta and Steinhart-Hart models on the same plot and not scaling



With the correct graphs, we can now see where the model is most accurate. Previously I used 0, 25, and 85 as the temperatures to get coefficients. There is a significant drop over the range we need, which is the 54-70 range. So instead of trying to get points over the entire B-accurate range, I should force the model to be accurate around the range I will actually be trying to measure!

The article I used previously found that 6-12% of consumers assessed the coffee as too cold over the range 56-68, so I'm going to pick our new points from 50 (definitively too cold), 70 (the most preferred average temp) and keep the 85 point.

$$\frac{1}{T} = A + B\ln(R) + C(\ln(R))^3$$

$$\frac{1}{50°C + 273.15K} = A + B\ln(3.671\times10^3\Omega) + C\ln(3.671\times10^3\Omega)^3$$

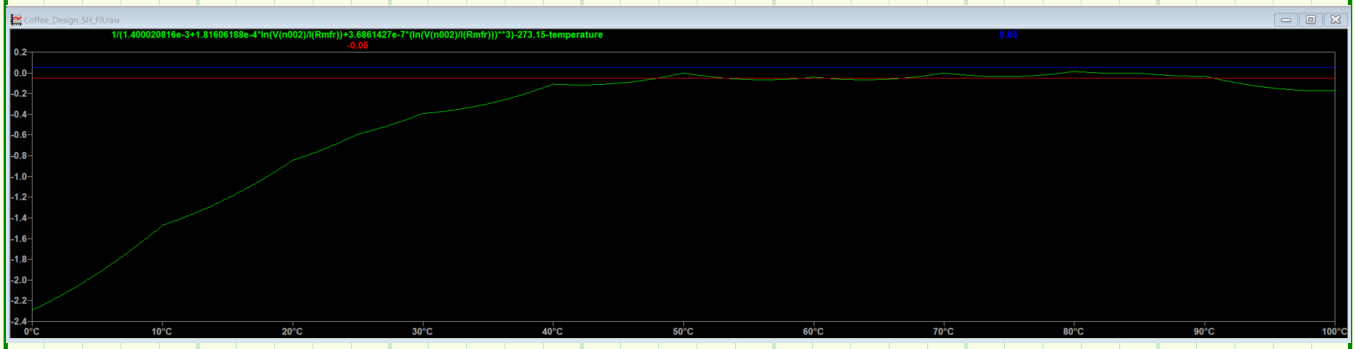
$$\frac{1}{70°C + 273.15K} = A + B\ln(1.783\times10^3\Omega) + C\ln(1.783\times10^3\Omega)^3$$

$$\frac{1}{85°C + 273.15K} = A + B\ln(1.071\times10^3\Omega) + C\ln(1.071\times10^3\Omega)^9$$

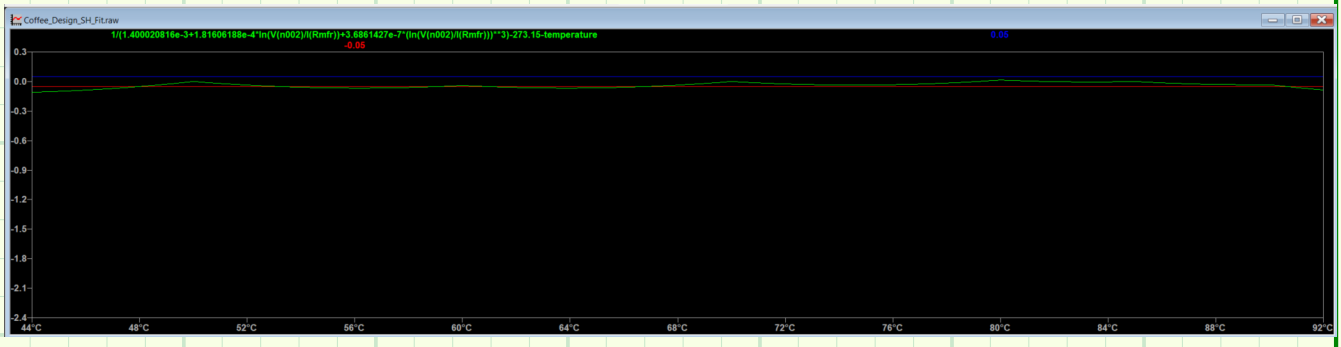
$A = 0.001420002081 6$

$B = 0.000181606188$

$C = 0.000000368614$

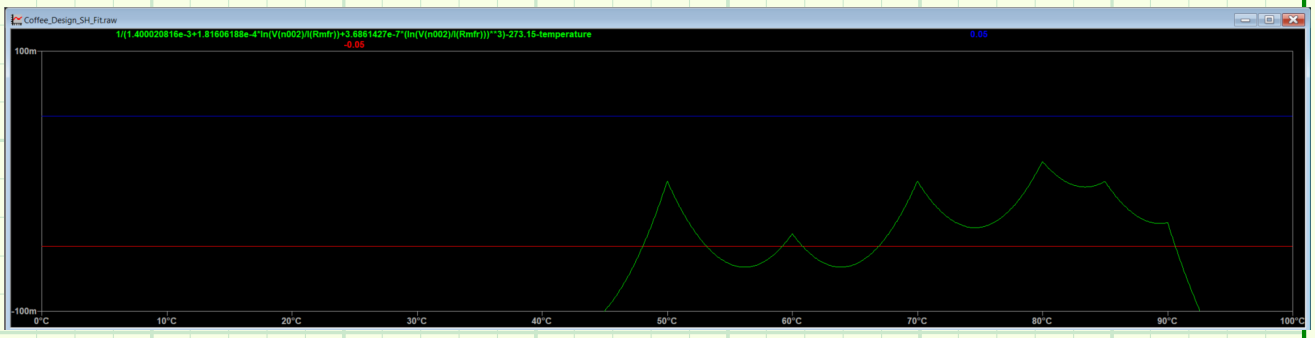


Zoomed to 50-85:

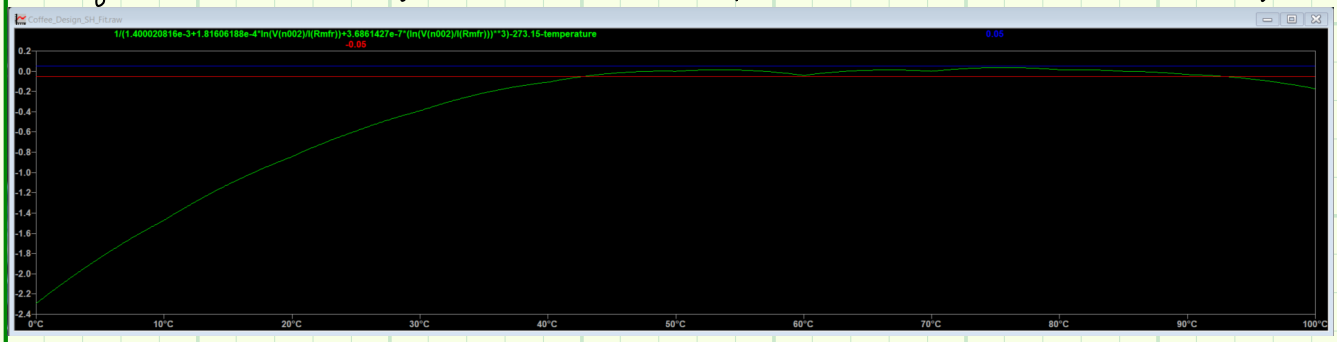


The range inside -0.05 to 0.05 is about 48°C - 53°C, 60°C, 68°C - 90°C. since 90°C is above, maybe we should refit inside 50°, 70°, and 80°C?

The worst dip occurs at ~55°C, where the error is 0.06603 below actual.



Have put in a request to Semitec to get 1°C separated table instead of interpolating the 10°C spaced one. Temporarily, we can interpolate Rmfr as shown, changing ln(R) against T to ln(R) against $\frac{1}{T}$ as well. I'm reporting the worse one in the table though.



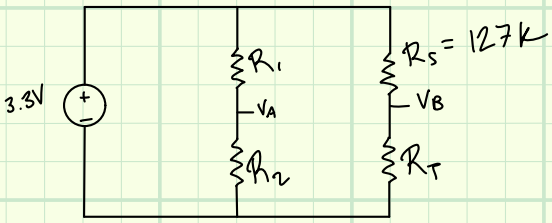
So far, worst model error is ~ -2.45°C at 0°C and at 100°, the error is around -0.2°C

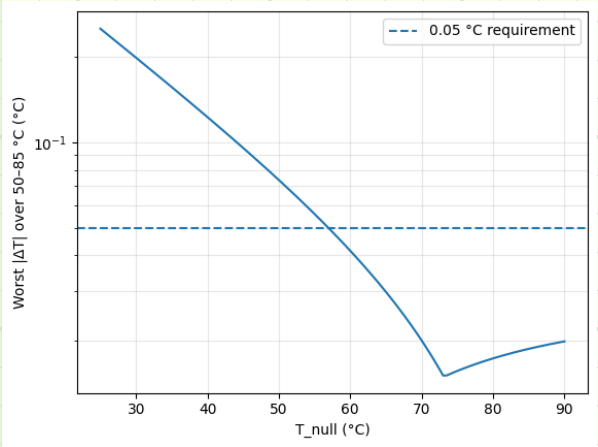
Power budget from A2 is 6 µW. We left the null at 25°C for the assignment, which would not serve us well. We should redesign around a higher T_{null}.

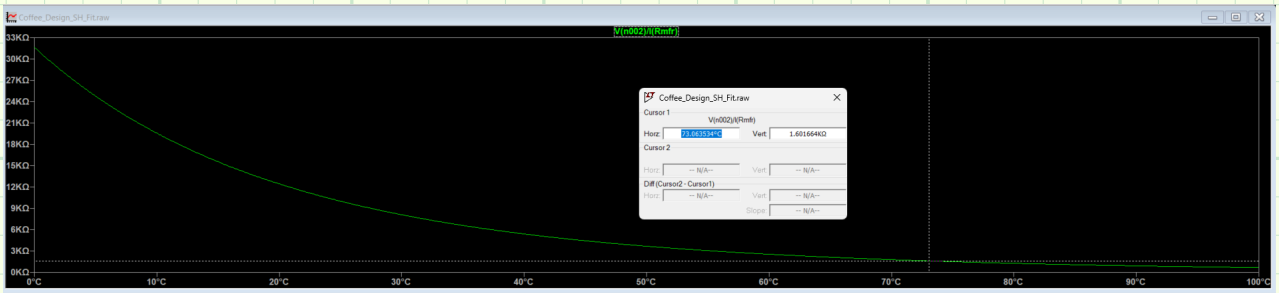
From Python script used in Assignment 1:

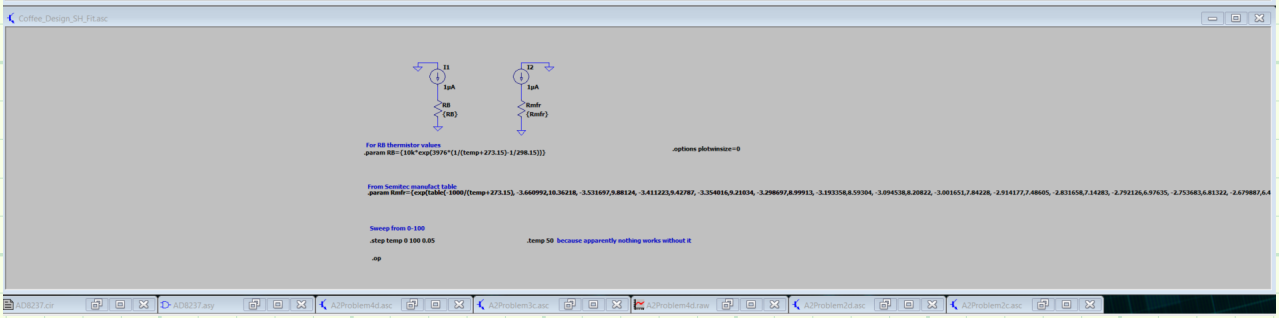
The optimal T_{null} would be 73°C at 0.0151 error within the range we used above, 50°C - 85°C.

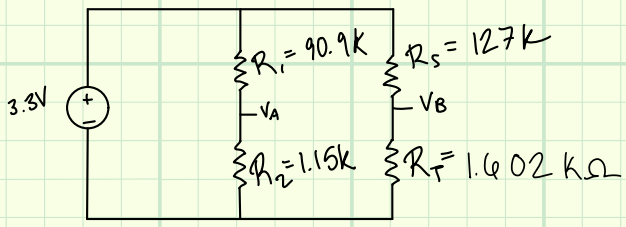
R_T at 73°C instead of 25°C, so not 10k anymore.





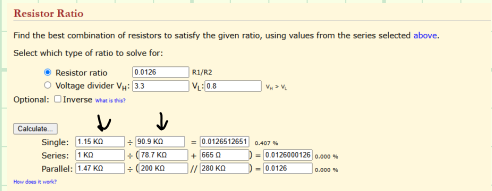






$$\frac{R_2}{R_1} = \frac{R_T}{R_S} = \frac{1602}{127{,}000}$$

$$\frac{R_2}{R_1} = 0.0126$$



Worst case power at 0°C and above 100°C

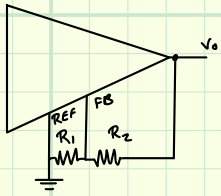


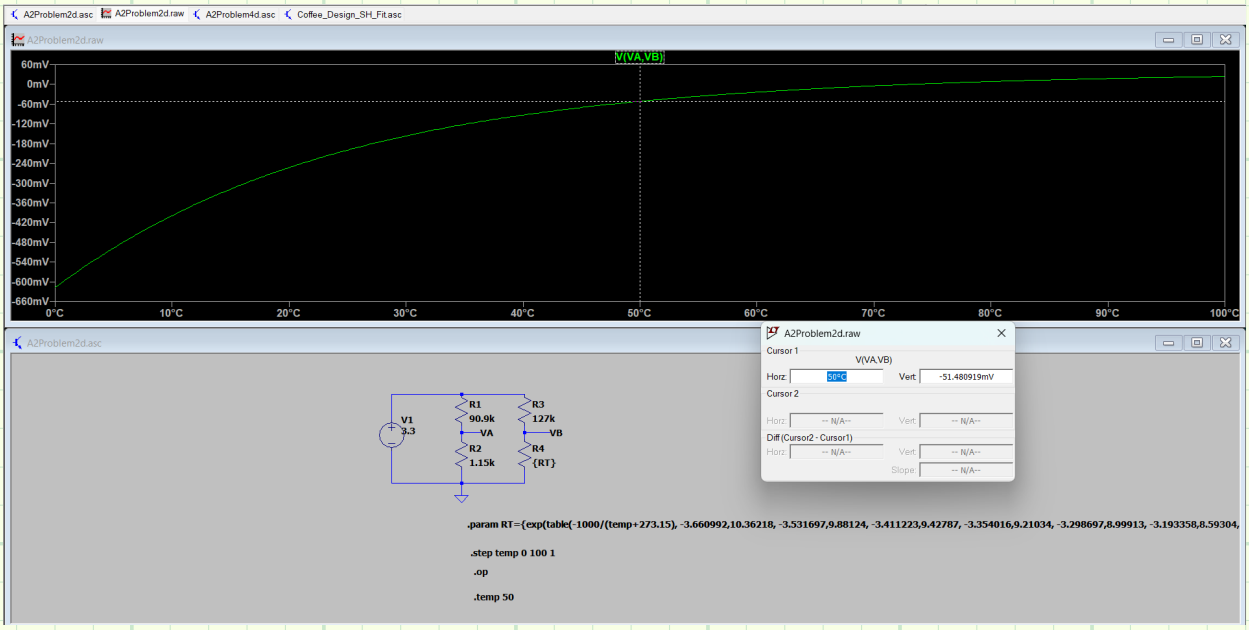
IA Use: AD8237.

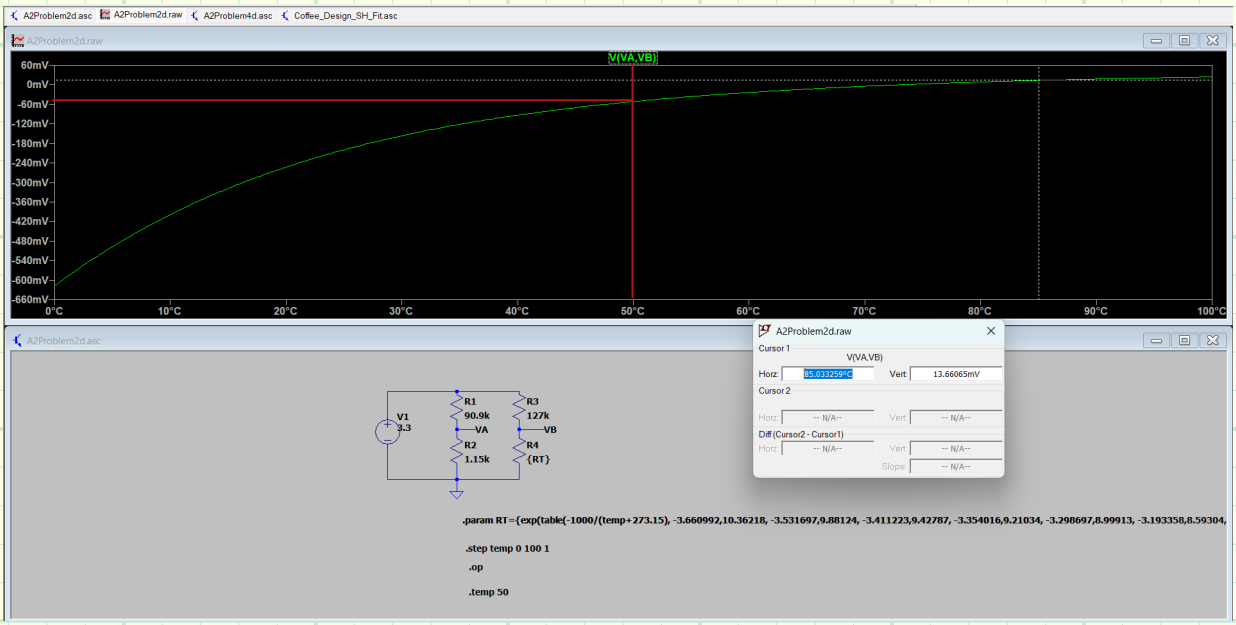
Datasheet

Recalculating gain: $V_{out} = V_{REF} + G(V_{in+} - V_{in-}) = V_{REF} + GV_{diff}$ $\quad G = 1 + \frac{R_F}{R_G}$

The range we are targeting is 50°C – 85°C, with the bridge nulled at 78°C. LTSpice reports V_a, V_b from −51.480919mV to ~13.660655mV.







The $V_{o,min}$ and $V_{o,max}$ are the same as in #2. 0.07V, and 3.23V

$$G_{max} = \frac{\Delta V_{all}}{\Delta V_{bridge}} = \frac{3.23 - 0.07\ V}{13.66065 - (-51.48091)} = \boxed{48.51}$$

$$1 + \frac{R_{FB}}{R_{REF}} \leq G_{max}$$

$$\frac{R_{FB}}{R_{REF}} \leq 47.51$$

low end

$$V_{REF} + G(-51.48091 \times 10^{-3} V) \leq 0.07V$$

high end

$$V_{REF} + G(13.66065V) \leq 3.23V$$

New gain: $1 + \frac{390k\Omega}{8.2k\Omega} = 48.5609$

This would put us above max gain though

Resistor Ratio

Find the best combination of resistors to satisfy the given ratio, using values from the series selected above.

Select which type of ratio to solve for:

Resistor ratio 47.51 R1/R2

Voltage divider V_H: 3.3 V_L: 0.8 $V_H > V_L$

Optional: Inverse What is this?

Calculate... RFB RREF

Single:	390 KΩ	÷	8.2 KΩ	=	47.56097560	0.107 %	
Series:	390 KΩ	÷	(8.2 KΩ	+	8.2 Ω)	=	47.51346214	0.007 %
Parallel:	120 KΩ	÷	(2.7 KΩ	//	39 KΩ)	=	47.52136752	0.024 %

How does it work?

Page 20 says that $R_{REF} || R_{FB} < 30k\Omega$.

SETTING THE GAIN

There are several ways to configure the AD8237. The transfer function of the AD8237 in the configuration in Figure 65 is

$$V_{OUT} = G(V_{+IN} - V_{-IN}) + V_{REF}$$

where:

$$G = 1 + \frac{R2}{R1}$$

Table 7. Suggested Resistors for Various Gains (1% Resistors)

R1 (kΩ)	R2 (kΩ)	Gain
None	Short	1.00
49.9	49.9	2.00
20	80.6	5.03
10	90.9	10.09
5	95.3	20.06
2	97.6	49.8
1	100	101
1	200	201
1	499	500
1	1000	1001

Whereas the ratio of R2 to R1 sets the gain, the designer determines the absolute value of the resistors. Larger values reduce power consumption and output loading; smaller values limit the FB input bias current and input impedance errors. If the parallel combination of R1 and R2 is greater than about 30 kΩ, the resistors start to contribute to the noise. For best output swing and linearity, keep $(R1 + R2) || R_L \geq 10\ k\Omega$.

If $R_{REF} = 1k\Omega$

$$1 + \frac{R_{FB}}{1k\Omega} < 48.51$$

$$R_{FB} < 47.51k\Omega$$

There's a 47.5k E96 resistor but I'll pick lower for tolerance and headroom

46.4k with 1% tol max is 46.864 and min is 45.936

$$G = 1 + \frac{46.864k}{1.01k\Omega} = 46.4 \leq 48.51 \checkmark$$

$$G = 1 + \frac{45.936k\Omega}{0.98k\Omega} = 47.87 \leq 48.51 \checkmark$$

Real Gain: $1 + \frac{46.4k}{1k}$

$= 47.4$

$V_o = V_{REF} + G V_{diff}$

low end

$V_{REF} + G(-51.48091 \times 10^{-3} V) \leq 0.07V$

$V_{REF} + 47.4(-51.48091 \times 10^{-3} V) \leq 0.07V$

$V_{REF} \leq 2.510195134$

high end

$V_{REF} + G(13.660665mV) \leq 3.23V$

$V_{REF} + 47.4(13.660665mV) \leq 3.23V$

$V_{REF} \leq 2.582499941$

We can make a voltage supply off our 3.3V that's powering the rest

$2.510195134 < V_{REF} < 2.582499941$



$V_{REF} = 2.55V$

V_{REF} window = 0.07154807V = 71.55mV

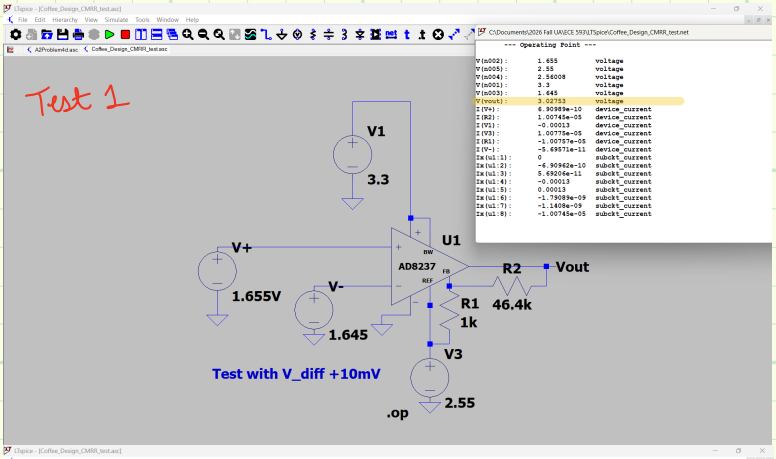
$$2.55 = 3.3 \frac{R_2}{R_1 + R_2}$$

$$0.772727 = \frac{R_2}{R_1 + R_2}$$ ← pick easy

$$0.7727 = \frac{10k}{R_1 + 10k}$$

$R_1 = 2.9412k\Omega$ nearest E96 is $2.94k\Omega$

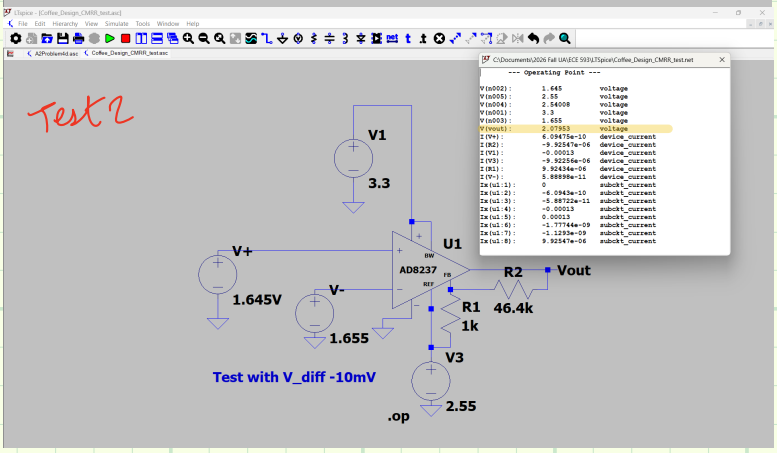
Using a voltage follower so the REF is real and not affecting the gain. Taken from Gage Rucker (fellow student)



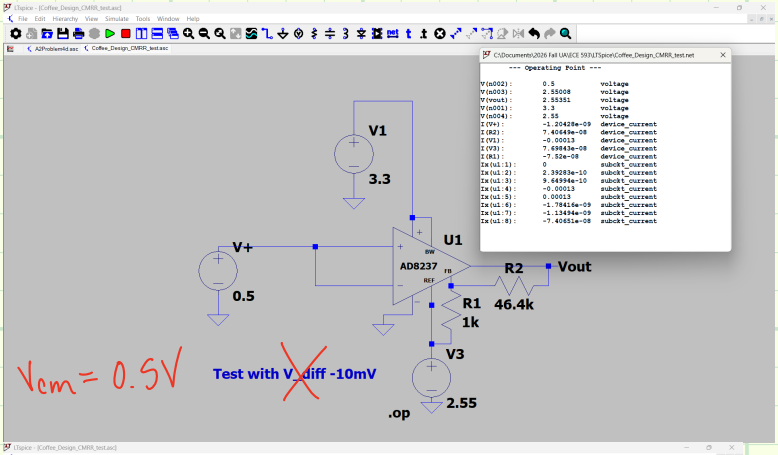
$$A_d = \frac{\Delta V_{out}}{\Delta V_{diff}}$$

$$A_d = \frac{3.02753 - 2.07953V}{10mV - (-10mV)}$$

$$A_d = 47.4$$



A_{cm}:

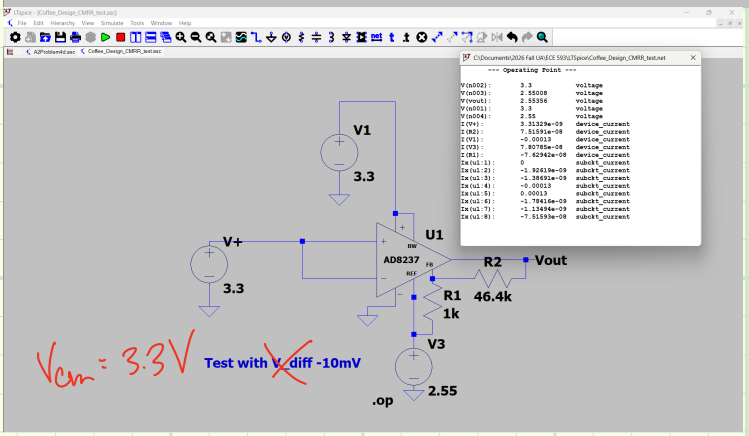


$$A_{cm} = \left| \frac{2.55351 - 2.55356}{3.3V - 0.5V} \right|$$

$$1.7857 \times 10^{-5}$$

$$CMRR = \frac{A_d}{A_{cm}}$$

$$\left| \frac{47.4}{1.7857 \times 10^{-5}} \right|$$



$$CMRR = 2.6544 \times 10^6$$

$$dB = 20\log(2.6544 \times 10^6)$$

$$dB = 128.479$$

within range ✓

$$\Delta T_{CMRR} = \frac{V_{em} / CMRR}{S_{bridge}} \approx \frac{0.034V / 10^{106/20}}{0.85\,mV/°C} \approx 0.0002°C$$

I want to power the board with a Pico pin, but pending instructor feedback. Using an assumed 3.7 LiPo battery right now. Worst case from battery is

Update: Investigating 3.3V nRF52480 output and noise:

Laptop 5V through USB is uncertain and I know how to compensate for battery error, but not individual C-ports. Maybe battery should still be J1 connector and a separate could power Seeed?

For filtering, I want to add decoupling capacitors like I have in previous circuit designs

Picking

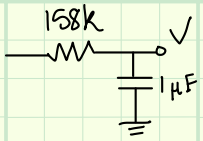
In notes, Sallen-Key used because INA333 gain limited by common-mode. AD8237 uses full-range, not limited by common-mode.

$K = 1$ because $G_{tot} = G_{supplied}$, $K = 1 + \frac{R_f}{R_a}$, no R_f or R_a

$Q = \frac{1}{3-1} = \frac{1}{2}$ $\quad f_{sig} = \frac{1}{2\pi(155)} = 10.6\,mHz$ using 1Hz like notes

$f_0 = \frac{1}{2\pi RC}$ ← 1μF

$1Hz = \frac{1}{2\pi R(C)}$



$R = 159.2k\Omega$ $\quad 158k\Omega$ E96

Power calcs:

$I_A = \frac{3.3}{90.9k + 1.15k} = 0.03585mA$ $\quad I_B = \frac{3.3}{127k + small}$

$I_B = 0.02598mA$

REF $I = \frac{3.3}{2.94k + 10k} = 0.255mA$

From voltage reference, total is $0.03585 + 0.02598 + 0.255 = 0.3169mA$

From R_{REF} and $R_{FB} = \frac{V_{outmax} - V_{REF}}{46.4k + 1k}$

$\frac{4.2V - 0.07 - 2.55}{46.4k + 1k} = 0.0333mA$

Total: $0.3169mA + 0.033mA + 550\mu A + 150\mu A + 28\mu A = 1.0779\,mA$

ADC selection:

Familiar with ADS1115 and it is I2C with multiple input pins. Can use 4.7K pull up resistors